

HUANXUAN LIAO

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🎓 EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 2023 – Now

PhD in Computer Science. In NLP, Supervisor: [Shizhu He](#) and [Kang Liu](#)

North China Electric Power University (NCEPU), Beijing 2019 – 2023

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Ranking 1/13

💡 RESEARCH INTERESTS

- **Efficient Long-Context Modeling** – long-context data synthesis, sparse architectures, and compression:
 - Community & survey: [Awesome-LCLM](#) (Stars > 2.2K); LCLM survey (core contributor)
 - Architecture & KV compression: [HyCo₂](#) (arXiv 2025), [Spark](#) (AAAI 2026)
 - Parametric RAG: [DyPRAG](#) (arXiv 2025)
 - Multimodal efficiency: [ResAdapt](#) (arXiv 2026)
- **Adaptive Model Learning** – turn instructions and knowledge into task-specific capacity:
 - Instruction / adapters: [TAGI](#) (NeurIPS 2024), [SHIP](#) (ACL 2026 Main)
 - Knowledge activation & distillation: [AAG](#) (COLING 2025), [SKIntern](#) (COLING 2025), [NesyCD](#) (AAAI 2025)
 - Continual adaptation: [SpaRTA](#) (ACL 2026 Main), [PaRSP](#) (ACL 2026 Main)
 - Policy optimization: [BuPO](#) (EMNLP 2026 Main)
- **Reliable Long-Horizon Agents** – persistent operational agency, tool use, evaluation, and agent memory:
 - Tool learning / eval / RL: [CITI](#) (AAAI 2025), [ETAPP](#) (ACL 2025 Main), [KATE](#) (ACL 2026 Findings), [Tool-RL-Box](#) (FAGEN@ICML 2026)
 - Operational agency: OPERATE
 - Agent memory & preference distillation: [Distill Yourself](#)

📖 SELECTED PUBLICATIONS [[FULL LIST](#)]

(* stands for equal contribution; Listed in reverse chronological order.)

- [Huanxuan Liao](#), Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, Kang Liu. [ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning](#). (arXiv 2026)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, Kang Liu. [Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs](#). (ACL 2026 Main, CCF-A)
- [Huanxuan Liao](#), Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, Kang Liu. [Spark: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning](#). (AAAI 2026, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Jun Zhao. [Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks](#). (AAAI 2025, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, Kang Liu. [SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models](#). (COLING 2025, CCF-B)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Shengping Liu, Jun Zhao. [Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering](#). (COLING 2025, CCF-B)
- [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, Jun Zhao. [From Instance Training to Instruction Learning: Task Adapters Generation from Instructions](#). (NeurIPS 2024, CCF-A)
- [Huanxuan Liao](#), Shizhu He, Yupu Hao, Jun Zhao, Kang Liu. [Beyond Hard and Soft: Hybrid Context Compression for Balancing Local and Global Information Retention](#). (arXiv 2025)
- Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, [Huanxuan Liao](#), et al. [A Comprehensive Survey on Long Context Language Modeling](#). (arXiv 2025, **Core Contributor**)

👤 INTERNSHIP/PROJECT EXPERIENCE

Tencent (Project UP) Beijing

May 2026 – Present

Intern Intern (Pretraining)

- **Sparse Attention Architecture:** Proposed **ROAM** for settings where Stage-1 selection/routing cost rivals or exceeds Stage-2 attention: replace per-token global routing with low-frequency Future Pool construction and high-frequency in-pool rerank, and co-design HBM residency with direct-paged Stage-2 to cut selection and KV-movement overhead.
- **Vision-Code Data Synthesis:** Built synthesis and conversion pipelines for games / SVG / sketch2code with license-aware ingest and behavioral admission, supporting multimodal pretraining data construction.
- **Multimodal Error Agent:** Built a multimodal error agent with an executable-assertion loop (Detect→first-causal localize→surgical repair→reverify) for process-level silent errors in synthetic data. Hard-oracle recall **0.798**; controlled SVG graph-CF Acc@5 **1.00** (vs flat replay 0.33); applied to pretraining data QC and synthetic-data feedback.

ByteDance Beijing

Jun. 2025 – May 2026

Intern Research Intern (Efficient Multimodal Understanding)

- **AIGC Detection Productionization:** Deployed efficient multimodal methods on the AIGC recognition line, achieving **>3%** online F1 gain and **30%–50%** compute savings, and extended to embedding retrieval / representation reuse.
- **Input-Side Resolution Adaptation:** Proposed **ResAdapt** + **CAPO** to allocate per-frame visual budget before encoding; discards **>90%** visual tokens, supports up to **16×** frames under fixed compute, and yields **>15%** relative gains on long-video reasoning.
- **Compression Benchmarking & Architecture Exploration:** Reproduced and compared multimodal visual-token / KV compression methods, and explored efficient variants (skip-layer, dynamic boundary compression) to widen the efficiency-accuracy Pareto frontier.
- **Operational Agency Benchmark:** Built **OPERATE** for persistent long-horizon operational agency: domain-native simulators evolve the environment, while the agent decides when to observe, intervene, or hold a standing plan; tasks are filtered by source lock and consumption proofs and scored with evidence-linked, same-seed counterfactuals.

Ant Group Beijing

Feb. 2025 – May 2025

Intern Research Intern (Long Context)

- **Hybrid Context Compression:** Proposed **HyCo₂** (soft global semantics + hard local pruning), achieving **+13.1%** relative EM on knowledge-intensive QA while cutting **~88.8%** tokens vs uncompressed.
- **Long-Context Survey:** Core contributor to the LCLM survey (data / architecture / evaluation) and maintainer of the Awesome long-context repository.

♡ AWARDS AND SCHOLARSHIPS

National Scholarship ×3
Beijing Outstanding Graduate

2021 / 2022 / 2025
2023.06

🔗 PROFESSIONAL SERVICES

Conference Reviewing: ACL ARR 2024–2026. NeurIPS 2025–2026. ICML 2026. COLM 2026. ICLR 2025–2026. AAAI 2026.

Community: [Awesome-LLM-Long-Context-Modeling](#) (GitHub Stars > 2.2K); LCLM-Horizon survey (core contributor); [NVIDIA/kvpress](#) (contributor); [Distill Yourself](#) (agent memory & preference distillation).